

## FLOODGAUGE · DESIGN DECISION RECORD

# FloodGauge recommendations accepted

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## DOCUMENT CONTROL

|             |                                                               |         |                |
|-------------|---------------------------------------------------------------|---------|----------------|
| Document ID | FLG-DDR-002                                                   | Version | 0.3            |
| Status      | <b>Draft</b>                                                  | Date    | 2026-10-02     |
| Author      | Amish Chadha                                                  | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                                 |         |                |
| Repository  | github.com/BoujeeEnjinia1701/floodgauge                       |         |                |
| Source      | docs/decisions/0002-recommendations-accepted.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                |
|---------|------------|--------------|-----------------------------------------------------------------------|
| 0.3     | 2026-10-02 | Amish Chadha | O1 and O3 decided by Amish on 2026-10-02 (FLG-DEC-001, items 5 and 6) |
| 0.2     | 2026-09-26 | Amish Chadha | Budget approved by Amish (\$160)                                      |
| 0.1     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002)                           |

# 0002: Recommendations accepted

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- **Date:** 2026-09-25
- **Status:** accepted. Every item below with a recommendation is "Decided by Amish, 2026-09-25: go with recommendation". Items without a recommendation stayed "Proposed, awaiting Amish" at this record; O1 and O3 were decided by Amish on 2026-10-02 (FLG-DEC-001, items 6 and 5): "i approve your recommendations for all 555 open decisions."

## ■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." At that point this repo had nine items adopted for TRL 3 work pending his review (FLG-DDR-001, D1 to D9), one open item with no recommendation (O1), and five further items raised by the TRL 3 calculations in docs/REVIEW.md (session 2026-09-25, TRL 3, "Still awaiting Amish", items 2 to 6). Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, and nothing here goes past TRL 3.

## ■ Options considered

The options for each item are those in docs/REVIEW.md (both 2026-09-25 sessions), FLG-DDR-001 and FLG-CAL-001 v0.1.

## ■ Decision

*Table 1. Items from FLG-DDR-001, now decided.*

| #  | Item              | Decision                                                                     | What changed in the repo                                                                            |
|----|-------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| D1 | Budget            | FieldNode costed in its own repo;<br>FloodGauge-specific parts \$150 or less | <code>budget_usd</code> stays \$150 (the recommended figure);<br>R16 wording as in FLG-REQ-001 v0.3 |
| D2 | Heads             | Street and drain heads on one node                                           | None beyond the status wording                                                                      |
| D3 | Street sensing    | 60 GHz radar; ultrasonic as the lower-cost variant                           | None beyond the status wording                                                                      |
| D4 | Drain head        | Ultrasonic in a slotted stilling tube with a wet probe                       | None beyond the status wording                                                                      |
| D5 | Head position     | Arm over the gutter, height to the clearance rule                            | Height now set under N2 below                                                                       |
| D6 | Alert bands       | 150 mm and 300 mm plus drain-full, set with the partner city                 | First band below 150 mm added under N5                                                              |
| D7 | Network           | Private or city gateway for events; TTN for pilots only                      | None beyond the status wording                                                                      |
| D8 | Privacy           | Levels only                                                                  | None beyond the status wording                                                                      |
| D9 | Pitch and problem | No change                                                                    | <code>project.yaml</code> and <code>README.md</code> keep the existing wording                      |

All nine: Decided by Amish, 2026-09-25: go with recommendation.

Table 2. Items raised at TRL 3, now decided.

| #  | Item                          | Decision                                                                                                                                                           | What changed in the repo                                                                                                                                                                                                                        |
|----|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N1 | R3 wording                    | Restate R3 as "continuous reading from 50 mm above the basin floor to 330 mm below the road, plus a drain-full signal above that"                                  | FLG-REQ-001 v0.4; R3 not met to met on paper                                                                                                                                                                                                    |
| N2 | Head height and R1            | Option (a): widen the R1 band to 2.5 to 5.0 m and set the height per site to the road authority's clearance rule, about 4.6 m where trucks use the curb lane       | R1 target in FLG-REQ-001 v0.4; lens 2.95 m to 4.6 m and pole 3.3 m to 4.9 m in <code>cad/src/model.py</code> ; FieldNode center 2.35 m to 3.0 m and street cable 2 m to 3 m to keep the I2C bus under 400 pF (350 pF); FLG-DWG-001 Rev P1 to P2 |
| N3 | R12 cable route               | Option (b) for pilots: out at the grate frame and along the curb face under a bolted steel cover; option (a), the conduit, at permanent sites laid with road works | BOM line 7 conduit (\$8.00) to surface cable cover (\$14.00); model item 7 redrawn; installation with no civil work, 120 min; R12 still not met on time                                                                                         |
| N4 | Arm anti-rotation             | Add a through-bolt or pinned clamp; the through-bolt is used                                                                                                       | M8 A4 through-bolt at the lower clamp in the model and BOM line 3 (\$18.00 to \$20.00); twist factor 1.45 to 37                                                                                                                                 |
| N5 | First alert band below 150 mm | The alert service supports a first band below 150 mm; its depth is set with the partner city under D6                                                              | FLG-PRC-001 v0.4 and FLG-REQ-001 v0.4 text; no depth chosen, because none was recommended                                                                                                                                                       |

All five: Decided by Amish, 2026-09-25: go with recommendation.

Table 3. Items still open.

| #  | Item                                                                                                                                            | Status                                                                                                                                                                                                                                                                                                                                                                              |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | Pilot partner and city for co-design and a first site. No preference stated and no recommendation made. The depth of the N5 band waits on this. | Decided by Amish, 2026-10-02 (FLG-DEC-001, item 6): a city or county flood agency with street flooding at grated curb inlets and an existing gauge or alert programme; first candidate type to approach, a county flood control district such as the Harris County Flood Control District; first alert band 100 mm unless the partner's practice says otherwise. Nothing is agreed. |
| 02 | R16 over budget after N3 and N4: \$153.50 against \$150 (new, raised by FLG-CAL-001 v0.2)                                                       | Decided by Amish, 2026-09-26: <code>budget_usd</code> \$160 (see below)                                                                                                                                                                                                                                                                                                             |
| 03 | R12 installation 120 min against 90 min on the surface route (new)                                                                              | Decided by Amish, 2026-10-02 (FLG-DEC-001, item 5): R12 relaxed to 120 min for the pilot surface route; 90 min stays the target for permanent sites on the conduit route.                                                                                                                                                                                                           |

## ■ Consequences

- FLG-REQ-001 v0.4, FLG-PRC-001 v0.4, FLG-CAL-001 v0.2 and FLG-DDR-001 v0.2 carry the new wording and numbers. FLG-DWG-001 is at Rev P2. The STEP, STL and media files are regenerated from the model.
- Requirement status: 2 not met (R12, R16), 3 at risk (R7, R9, R14), 1 not verifiable at TRL 3 (R10), 8 met on paper, 3 met by design. R3 moves from not met to met on paper; R16 moves from met on paper to not met.
- Cost: FloodGauge-specific parts \$145.50 to \$153.50; complete gauge \$271.50 to \$279.50. `budget_usd` stays \$150.
- Mass on the pole 4.62 kg to 4.74 kg; still under 5.0 kg.
- A site now needs a pole about 4.9 m tall above the sidewalk where tall vehicles use the curb lane, and a mobile elevating platform to fit the arm.
- Cross-repo actions are listed in `docs/REVIEW.md` ; no other repo was edited.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building, testing or purchasing.

## ■ Budget approved, 2026-09-26

On 2026-09-26 Amish wrote, in chat: "i approve all the budget items."

- Budget set to \$160 to cover the priced BOM: decided by Amish, 2026-09-26. This settles O2. FloodGauge-specific parts are \$153.50 (lines 2 to 10; the \$126.00 FieldNode core stays costed in its own repo under D1), so R16 moves from not met to met on paper with a \$6.50 margin.
- Requirement status: 1 not met (R12), 3 at risk, 1 not verifiable at TRL 3, 9 met on paper, 3 met by design.
- Files changed: `project.yaml` ( `budget_usd` 150 to 160); FLG-REQ-001 v0.5; FLG-CAL-001 v0.3, `docs/04-calcs/sizing.py` (target read from `budget_usd` ) and `results.csv` ; FLG-PRB-001 v0.5 and FLG-PRC-001 v0.5 (budget figure); `README.md` ; `bom/bom-notes.md` ; `docs/REVIEW.md` .